Customer Behavior Analysis Report

Project Title:

Customer Behavior Analysis Using RFM Analysis and K-Means Clustering

Introduction:

Customer behavior analysis helps businesses understand purchasing patterns, customer engagement, and retention trends. In this project, customer transaction data was analyzed to identify customer segments, purchase behavior, and churn risks.

The project uses RFM (Recency, Frequency, Monetary) analysis and K-Means clustering to group customers based on their purchasing activities.

The analysis helps Alfido Tech improve customer engagement, retention, and marketing strategies.

Objectives:

The main objectives of this project are:

1. Perform data cleaning and feature engineering.
2. Analyze customer purchasing behavior.
3. Segment customers using RFM analysis and clustering.
4. Visualize purchasing and retention trends.
5. Identify churn-risk customers.
6. Provide actionable recommendations for Alfido Tech.

Dataset Description:

The dataset contains customer transaction and demographic information.

Dataset Columns

- Customer ID
- Purchase Date
- Product Category
- Product Price
- Quantity
- Total Purchase Amount
- Payment Method
- Customer Age
- Returns
- Customer Name
- Age
- Gender
- Churn

Technologies Used:

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Data Cleaning:

The following data cleaning steps were performed:

1. Removed duplicate rows.
2. Removed missing values.
3. Converted the Purchase Date column into datetime format.
4. Verified column data types.

Code Used

```
# Check missing values
print(df.isnull().sum())
# Remove duplicate rows
df.drop_duplicates(inplace=True)
# Remove missing values
df.dropna(inplace=True)
# Convert Purchase Date into datetime format
df['Purchase Date'] = pd.to_datetime(df['Purchase Date'])
# Verify dataset
print(df.info())
```

➢ Feature Engineering:

RFM features were created for customer segmentation.

➢ **RFM Meaning**

- Recency: Days since last purchase
- Frequency: Number of purchases
- Monetary: Total purchase amount

RFM Table Creation

```
# Find latest purchase date
latest_date = df['Purchase Date'].max()
# Create RFM values
rfm = df.groupby('Customer ID').agg({
  'Purchase Date': lambda x: (latest_date - x.max()).days,
  'Customer ID': 'count',
  'Total Purchase Amount': 'sum'
})
# Rename columns
rfm.columns = ['Recency', 'Frequency', 'Monetary']
# Display first rows
print(rfm.head())
```

➢ Customer Segmentation:

K-Means clustering was used to segment customers into different groups.

➢ **Clustering Process**

1. Normalize RFM values.
2. Apply K-Means clustering.
3. Assign cluster labels.
4. Analyze cluster characteristics.

Customer Segment Profiles:

Cluster 0 – Loyal Customers

- High purchase frequency
- Moderate to high spending
- Regular purchasing behavior

Cluster 1 – High Value Customers

- Highest spending customers
- Premium buyers
- Important revenue contributors

Cluster 2 – At-Risk Customers

- Long time since last purchase
- Lower engagement
- Higher churn probability

Cluster 3 – New or Occasional Customers

- Lower purchase frequency
- Recent customers
- Potential future loyal customers

Visualizations and Analysis:

1. **Monthly Purchase Trend:**

- This visualization shows monthly sales and purchasing trends.

Observation

- Some months showed higher sales activity, indicating seasonal purchasing behavior.

2. **Customer Segment Visualization:**

- Scatter plots were used to visualize customer clusters based on frequency and monetary values.

Observation

- Distinct customer groups were identified using clustering.

3. **Product Category Analysis:**

- Product category analysis identified the most purchased product types.

Observation

- Certain product categories generated significantly higher customer engagement.

4. **Gender-wise Purchase Analysis:**

- Purchasing behavior was analyzed based on gender.

Observation

- Different customer groups showed varying purchasing behaviors.

5. **Churn Analysis:**
The churn column was analyzed to understand retention trends.

Observation

- Some customers showed high churn probability, indicating retention challenges.

➢ Key Findings

1. A small group of customers contributed significantly to revenue.
2. Some customers purchased frequently but spent less.
3. Seasonal purchase trends were identified.
4. Certain product categories performed better than others.
5. Churned customers showed lower purchasing frequency.
6. Customer segmentation improved understanding of purchasing behavior.

➢ Actionable Recommendations for Alfido Tech

1. **Implement Loyalty Programs**
Reward frequent customers with:
    - Loyalty points
    - Cashback offers
    - Exclusive discounts

    This can improve customer retention.

2. **Re-engage Churn-Risk Customers**
Send:
    - Personalized emails
    - Discount offers
    - Product recommendations

    This can reduce customer churn.

3. **Focus on High-Performing Product Categories**
Promote top-selling product categories through:
    - Advertisements
    - Featured listings
    - Seasonal promotions

    This can increase revenue.

4. **Use Personalized Marketing**
Analyze:
    - Gender
    - Age
    - Purchase behavior

    to recommend suitable products.
    This can improve customer engagement.

5. **Introduce Retention Campaigns**
Offer:

- Membership plans
- Festival discounts
- Referral rewards
- Limited-time offers

This encourages repeat purchases.

Conclusion

This project successfully analyzed customer behavior using RFM analysis and K-Means clustering techniques. Different customer segments were identified based on purchasing patterns, spending behavior, and customer activity.
The project also highlighted churn trends and purchasing behavior insights that can help Alfido Tech improve customer engagement and retention strategies.
The analysis demonstrates how data analytics and machine learning techniques can support better business decision-making.

Future Improvements

Future enhancements may include:

1. Predictive churn analysis using machine learning models.
2. Real-time customer recommendation systems.
3. Interactive dashboards using Power BI or Tableau.
4. Advanced customer lifetime value prediction.
5. Deep learning-based customer behavior prediction.

Visualizations and Code Snippets:

Monthly Purchase Trend

```
monthly_sales = df.groupby(
df['Purchase Date'].dt.month
)['Total Purchase Amount'].sum()
plt.figure(figsize=(8,5))
monthly_sales.plot(marker='o')
plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.show()
```

Visualization Output

- Displays monthly sales trends.
- Helps identify seasonal purchasing behavior.

Customer Segment Visualization

```
plt.figure(figsize=(8,6))
sns.scatterplot(
x=rfm['Frequency'],
y=rfm['Monetary'],
hue=rfm['Cluster'],
palette='Set2'
)
plt.title("Customer Segments")
plt.xlabel("Frequency")
plt.ylabel("Monetary")
plt.show()
```

Visualization Output

- Shows customer clusters.
- Identifies loyal and high-value customers.

Product Category Pie Chart

```
plt.figure(figsize=(8,8))
sales_data = df.groupby('Product Category')[
'Total Purchase Amount'
].sum()
plt.pie(
sales_data,
labels=sales_data.index,
autopct='%1.1f%%',
startangle=90
)
plt.title("Sales by Product Category")
plt.show()
```

Visualization Output

- Shows sales contribution of each product category.
- Identifies top-performing product groups.

```
# Estimated Profit
df['Profit'] = df['Total Purchase Amount'] * 0.20
profit_data = df.groupby('Product Category')[
'Profit'
].sum()
plt.figure(figsize=(8,8))
plt.pie(
profit_data,
labels=profit_data.index,
autopct='%1.1f%%',
startangle=90
)
plt.title("Profit by Product Category")
plt.show()
```

Visualization Output

- Displays estimated profit contribution by category.
- Helps identify profitable categories.

Quantity by Customer Name

```
quantity_data = df.groupby('Customer Name')[
'Quantity'
].sum().sort_values(ascending=False).head(10)
plt.figure(figsize=(12,6))
quantity_data.plot(kind='bar')
plt.title("Top 10 Customers by Quantity Purchased")
plt.xlabel("Customer Name")
plt.ylabel("Total Quantity")
plt.xticks(rotation=45)
plt.show()
```

Visualization Output

- Identifies customers purchasing the highest quantity.
- Useful for loyalty analysis.

Churn Analysis

```
plt.figure(figsize=(6,4))
sns.countplot(
x='Churn',
data=df
)
plt.title("Churn Analysis")
plt.show()
```

Visualization Output

- Displays churn distribution.
- Helps identify retention challenges.

Correlation Heatmap

```
plt.figure(figsize=(8,5))
sns.heatmap(
rfm[['Recency', 'Frequency', 'Monetary']].corr(),
annot=True,
cmap='coolwarm'
)
plt.title("Correlation Heatmap")
plt.show()
```

Visualization Output

- Shows relationships between Recency, Frequency, and Monetary values.
- Useful for customer behavior analysis.

Final Summary

This project successfully analyzed customer behavior using data analytics and machine learning techniques.
The project performed:

- Data cleaning and preprocessing
- Feature engineering using RFM analysis
- Customer segmentation using K-Means clustering
- Purchase trend visualization

- Churn and retention analysis
- Product category analysis
- Business recommendation generation

The analysis helped identify loyal customers, high-value customers, and churn-risk customers.
The findings and recommendations can help Alfido Tech improve customer engagement, retention, and sales performance.

References of Dataset
https://www.kaggle.com/datasets/bhanupratapbiswas/customer-behavior-analysis